

Affective-CoT: Decomposing Multimodal Emotion Reasoning through a Hierarchical Cognitive Workflow

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Abstract

Multimodal Emotion Recognition (MER) has advanced significantly with the advent of Multimodal Large Language Models (MLLMs), which enable generative, descriptive understanding of complex human affect. However, the inherent "black-box" nature of these end-to-end models limits their trustworthiness and applicability in high-stakes domains, particularly due to their opacity in handling conflicting cross-modal cues (e.g., sarcasm). To address this critical gap, we propose Affective-CoT, a novel hierarchical framework that explicitly decouples perception from reasoning to achieve interpretable and faithful emotion analysis. Our framework utilizes specialized perception models to extract structured semantic evidence from raw audiovisual streams, which is then integrated and arbitrated by a central reasoning LLM executing a meticulously designed Cognitive Workflow. Critically, Affective-CoT generates a nuanced emotion description grounded in a transparent, human-interpretable reasoning trace. The efficacy of our framework was decisively validated by securing first place in the official MER-2025 Descriptive Emotion Understanding (DES) challenge. This result not only highlights the superiority of our method but also champions a new paradigm for building scrutable and trustworthy affective computing systems.

CCS Concepts

• **Human-centered computing** → *HCI design and evaluation methods*; • **Computing methodologies** → *Multi-agent systems*.

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Keywords

Open-Vocabulary Emotion Understanding, Multimodal Large Language Models, Chain-of-Thought, Multimodal Emotion Recognition

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1 Introduction

To transition Multimodal Emotion Recognition (MER) from academic benchmarks to high-stakes applications—such as mental health diagnostics, adaptive educational systems, or even judicial sentiment analysis—a system's decision-making must be not only accurate but also fundamentally trustworthy and auditable [36]. However, a critical paradox is emerging in the field. The very technologies driving the latest performance leap, namely Multimodal Large Language Models (MLLMs) [16, 19, 40, 42], are simultaneously deepening this "trust deficit" due to their inherent black-box nature. While this new paradigm has shifted the field from simple classification to nuanced, generative understanding [30], and has produced models like AffectGPT [27] and Emotion-LLaMA [8] capable of generating remarkably fluent descriptions, their opaque processes obscure the very reasoning pathways essential for establishing trust. This creates a fundamental barrier to their adoption in domains where the 'why' of a decision is as critical as the 'what'.

Despite their superior performance, the inherent "black-box" nature of these models introduces severe challenges [12]. Their internal mechanisms for weighing evidence and resolving conflicts across modalities are opaque. When a person's facial expression (e.g., a smile) contradicts their verbal content (e.g., negative words) or acoustic features (e.g., a trembling voice)—as seen in complex emotions like sarcasm or forced smiles—current models often fail to provide a reliable basis for their judgment (see Figure 1 for an example), a recognized open challenge in the field [1]. This opacity creates a fundamental barrier to their adoption in domains where the

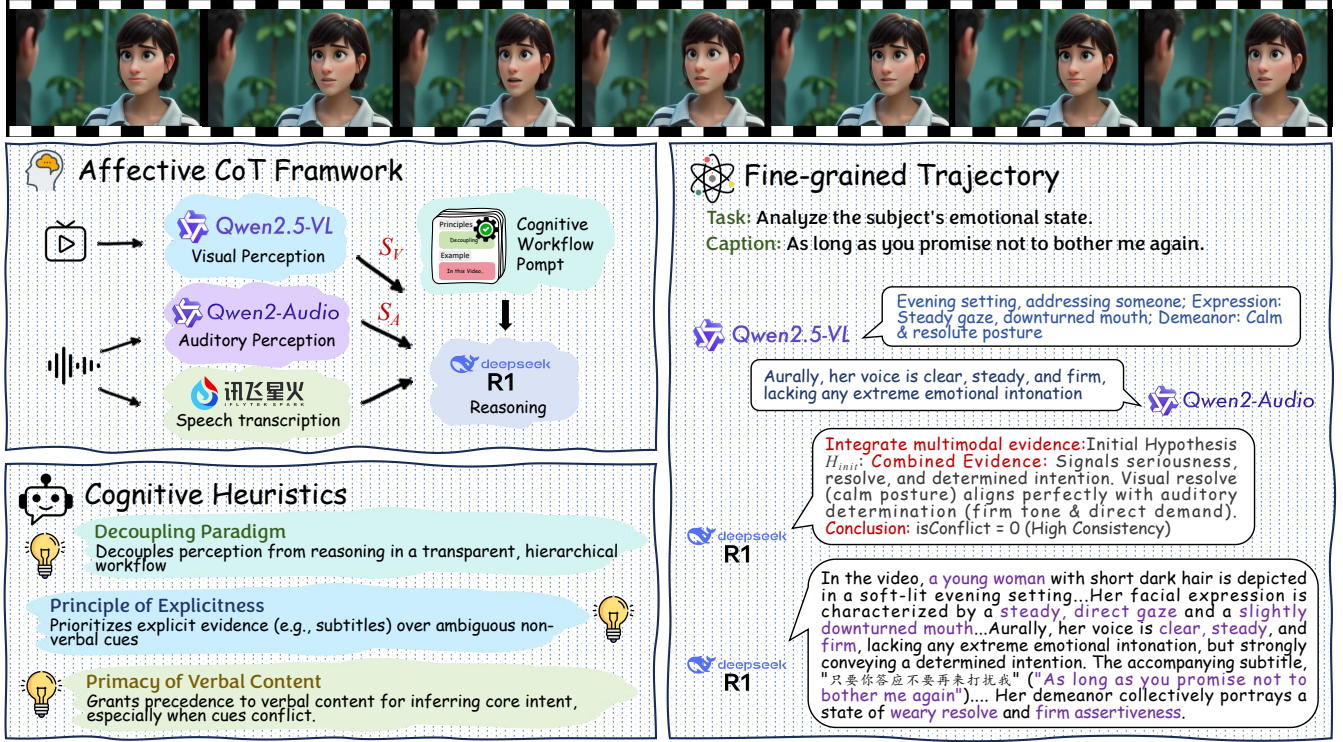


Figure 1: This figure illustrates the workflow of our Affective-CoT framework, which first employs specialized models like Qwen2.5-VL and Qwen2.5-Audio to encode raw multimodal inputs into structured evidence. This evidence is then fed into a central Cognitive Workflow that analyzes and resolves any cross-modal conflicts, forming a core emotional hypothesis. Finally, a reasoning LLM uses this hypothesis and the complete reasoning trace to generate a final, evidence-grounded explanation of the subject’s emotion.

‘why’ of a decision is as critical as the ‘what’. While our proposed hierarchical approach is inherently more computationally intensive than end-to-end models, we argue this is a necessary trade-off. In high-stakes applications like mental health diagnostics or judicial sentiment analysis, the demand for transparent, auditable, and trustworthy reasoning far outweighs the need for raw processing speed. Our work deliberately prioritizes this scrutable paradigm.[13]. Furthermore, the reasoning process is implicitly entangled within the model’s billions of parameters, making it impossible to trace the decision path for effective failure analysis. This results in plausible “post-hoc explanations” that are not faithful to the actual reasoning process [37].

To address these challenges and advance MER toward greater trustworthiness and interpretability, this paper introduces Affective-CoT (Affective Chain-of-Thought), a novel hierarchical cognitive framework. Inspired by the “perception-analysis-synthesis” cognitive process of human experts and the Chain-of-Thought (CoT) technique in LLMs [43], the core idea of Affective-CoT is to explicitly decouple perception from reasoning. It first employs specialized unimodal perception models to transform raw, unstructured audio-visual data streams into structured, semantic-rich “evidence.” Subsequently, a central reasoning engine follows a predefined “cognitive workflow” to integrate this evidence, perform consistency checks, arbitrate conflicts, and ultimately formulate a well-supported emotional judgment and explanation. This design not only renders

the entire decision-making process transparent and traceable but also significantly enhances the model’s robustness and accuracy in handling complex and ambiguous emotions.

The main contributions of this paper are threefold:

- We propose Affective-CoT, a novel hierarchical framework for interpretable MER that transforms the opaque end-to-end generation task into a structured and controllable process by decoupling perception from reasoning.
- We design a Cognitive Workflow with heuristic protocols that guides an LLM to emulate human-expert reasoning.
- Extensive experiments on the challenging MER-2025 benchmark demonstrate that our method achieves highly interpretability and faithfulness of the generated descriptions.

2 Related Work

2.1 Multimodal Emotion Recognition (MER)

MER aims to recognize human emotions by fusing information from multiple modalities, such as vision[23], audio[35], and text[11]. Early research primarily focused on feature engineering and classical fusion strategies, including early (feature-level[6]) and late (decision-level[4, 34]) fusion. With the advent of deep learning, attention-based[17, 41] models like Transformers have been widely applied, significantly boosting performance on benchmark datasets

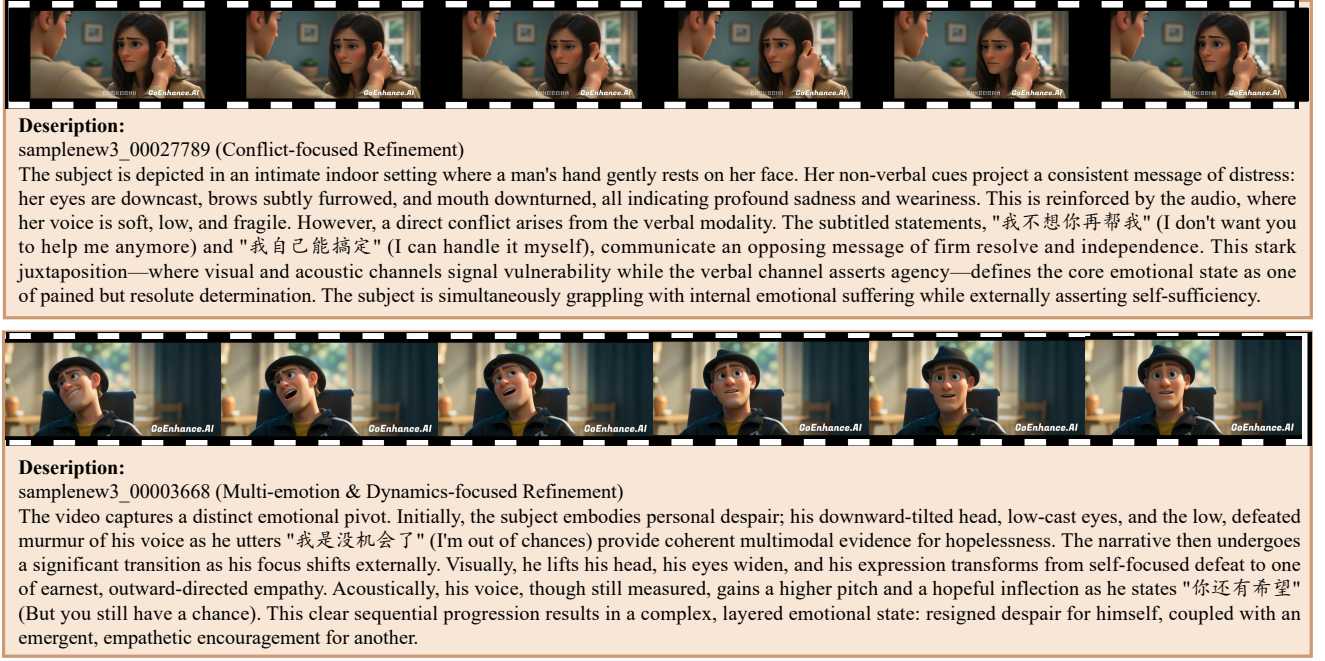


Figure 2: Case studies of the Affective-CoT framework reasoning on complex emotions. (a) Coexistence illustrates the framework’s ability to resolve conflicting cues, showing how contradictory emotions like sadness and determination can coexist. (b) Diversity highlights the framework’s capacity to track a diversity of emotions over time, identifying a clear transition from a state of despair to one of empathy

such as IEMOCAP [5] and CMU-MOSEI [45] by dynamically learning complex inter-modal alignments.

More recently, the emergence of Multimodal Large Language Models (MLLMs) has ushered MER into a new era. Foundational models like Video-LLaMa [46] and Qwen2.5-VL [3] have demonstrated powerful video understanding capabilities. Building on this, works such as AffectGPT [27] and OV-MER [25] have fine-tuned these general-purpose models for emotion description, enabling a leap from closed-vocabulary classification to open-vocabulary generation. However, these models predominantly follow an end-to-end paradigm; despite achieving state-of-the-art performance, their internal decision logic remains a “black box.” In contrast, Affective-CoT does not seek to design a larger or more intricate end-to-end model. Instead, we propose a new, transparent, and sequential reasoning paradigm to orchestrate existing large models, shifting the focus from what they do to how and why they do it.

2.2 Explainable MER in MLLMs

Model interpretability is crucial for establishing user trust and enabling the adoption of technology in critical domains. In MER, early interpretability efforts centered on visualizing attention weights, purporting to show which features or modalities the model “looked at” during decision-making [15]. However, whether attention faithfully reflects model rationale remains a subject of debate [20].

With the popularization of LLMs, leveraging their generative capabilities to produce post-hoc explanations for their own decisions has become a new trend. Some approaches prompt the model to

generate an explanatory text alongside its answer [26]. While these methods offer a degree of transparency [18], the resulting explanations are often “rationalizations” rather than genuine “reasoning.” The model first makes a judgment and then constructs a plausible justification, which fails to guarantee faithfulness. Our Affective-CoT framework, through its intrinsic causal structure—performing explicit evidence analysis and conflict arbitration before generating a final judgment—aims to fundamentally address this issue, ensuring the explanation is a faithful reflection of the model’s actual decision process.

2.3 Chain-of-Thought and Reasoning

The introduction of Chain-of-Thought (CoT) [43] demonstrated that prompting an LLM to generate intermediate reasoning steps can significantly improve its performance on complex tasks like mathematical and commonsense reasoning. This concept has rapidly evolved into a more general paradigm of “decomposed reasoning,” where a complex task is broken down into a series of simpler, LLM-executable sub-tasks. The ReAct framework [44], for example, integrates reasoning with actions, enabling an LLM to interact with external tools to accomplish a goal.

Inspired by these works, we are the first to systematically introduce the principles of CoT and decomposed reasoning to the domain of multimodal emotion understanding. Existing CoT research has predominantly focused on textual or structured data, with little work on processing and arbitrating unstructured and potentially conflicting perceptual information from different modalities. Affective-CoT extends this paradigm by designing a “cognitive

workflow” that is not merely a sequence of steps but a complex reasoning chain incorporating evidence weighting and conflict resolution rules, simulating an expert’s analytical process. This represents a significant extension and refinement of the CoT principle in a novel and challenging application domain.

3 Methodology

To address the inherent challenges of “black-box” reasoning and conflict resolution in multimodal emotion understanding, we propose Affective-CoT (Affective Chain-of-Thought), a novel hierarchical framework that emulates a human expert’s cognitive workflow. Our approach explicitly decouples low-level perceptual processing from high-level cognitive reasoning, thereby enhancing the model’s interpretability, faithfulness, and accuracy, particularly for complex emotional states. The core of Affective-CoT is not a complex pipeline of multiple, independent model calls, but rather a **single, comprehensive reasoning process executed by a central Reasoning LLM ($R - LLM$), orchestrated by a meticulously designed master prompt**. This prompt enforces a structured, multi-step cognitive workflow, transforming raw multimodal data into a final, interpretable analysis. To ensure full reproducibility, all meticulously designed prompts and our source code will be made publicly available at <https://github.com/GPNU-AIoT/Affective-CoT>.

3.1 Framework Overview

The design of Affective-CoT is grounded in the principles of hierarchical information processing, a cornerstone of cognitive science, and the decomposed reasoning paradigm demonstrated by Chain-of-Thought (CoT) prompting in Large Language Models [43]. As depicted in Figure 1, the framework conceptually operates in three sequential stages: Multimodal Evidence Perception, Cognitive Workflow Reasoning, and Descriptive Synthesis. Given a multimodal input segment $X = (V, A, T)$, where V represents the video stream, A the audio stream, and T the transcribed text, our objective is to generate a descriptive emotion label E_{desc} and a faithful explanation $Expl$. The overall function of our framework, F_{CoT} , can be expressed as:

$$(E_{desc}, Expl) = F_{CoT}(V, A, T) \quad (1)$$

This function is realized through the coordinated operation of the three distinct stages, which we detail in the subsequent sections.

3.2 Stage 1: Multimodal Evidence Perception

The initial stage of our framework is dedicated to perception, where specialized multimodal models act as domain experts to extract structured, semantic evidence from their respective modalities. This modular design allows for leveraging the unique strengths of different state-of-the-art models for specific perceptual tasks. The Visual Semantic Extractor (f_{VLLM}) employs a powerful Video Large Language Model, such as Qwen2.5-VL [3], to process the raw video stream V . Guided by a meticulously designed prompt P_V , the extractor’s operation is formalized as $S_V = f_{VLLM}(V, P_V)$. The prompt P_V is engineered to guide the model to output a structured JSON representation, S_V , which details primary subject cues, such as dominant facial expressions and gestures; contextual elements, including the environment and the behavior of other individuals; and

Algorithm 1 Cognitive Workflow for Affective Reasoning

Require: Structured evidence S_V, S_A ; Master prompt P_R .

Ensure: Core emotional hypothesis H_{emo} ; Reasoning trace R_{trace} .

```

1: function COGNITIVETWORKFLOW( $S_V, S_A, P_R$ )
   Step 1: Evidence Integration & Initial Assessment
2:    $E_{int} \leftarrow \text{Synthesize}(S_V, S_A)$ 
3:    $H_{init} \leftarrow \text{GenerateHypothesis}(E_{int}, P_R)$   $\triangleright$  LLM-based
4:    $R_{trace} \leftarrow \text{InitializeTrace}(H_{init})$ 
   Step 2: Cross-Modal Consistency Analysis
5:    $isConflict, C_{points} \leftarrow \text{AnalyzeConsistency}(E_{int}, P_R)$ 
6:   if  $isConflict$  then
7:      $R_{trace} \leftarrow \text{AppendToTrace}(R_{trace}, C_{points})$ 
    $\hookrightarrow$  Step 3: Conflict Resolution
8:      $H_{res} \leftarrow \text{ResolveConflict}(C_{points}, P_R)$ 
9:      $H_{emo} \leftarrow H_{res}$ 
10:     $R_{trace} \leftarrow \text{AppendToTrace}(R_{trace}, H_{res})$ 
11:   else
12:      $H_{emo} \leftarrow H_{init}$ 
13:   end if
14:   return  $H_{emo}, R_{trace}$ 
15: end function

```

temporally localized significant events with associated confidence scores.

Concurrently, the Auditory Semantic Extractor (f_{ALLM}) utilizes an Audio Large Language Model, for instance, Qwen2-Audio [10], to analyze the audio stream A and its corresponding text transcript T . This process, defined by $S_A = f_{ALLM}(A, T, P_A)$, yields a structured JSON object S_A . This object encapsulates key vocal characteristics like prosody and perceived tone, a semantic analysis of the verbal content, and significant non-verbal auditory events such as sighs or laughter, complete with timestamps. The use of a structured format for both S_V and S_A is a critical design choice, transforming unstructured data into a machine-readable format amenable to the symbolic reasoning in the subsequent stage.

3.3 Stage 2: Cognitive Workflow Reasoning

This stage forms the cognitive core of Affective-CoT, executed by a central Reasoning LLM ($R - LLM$), for which models like Llama-3 [14] are well-suited. The $R - LLM$ is steered by a master prompt, P_R , which enforces a strict cognitive workflow on the aggregated evidence $S_{VA} = \{S_V, S_A\}$. The workflow, formalized in Algorithm 1, methodically guides the model through a series of analytical steps.

Central to this workflow is a **Domain-General Heuristics Protocol** embedded within P_R , which provides explicit rules for adjudicating conflicting evidence. The *Principle of Explicitness* dictates that explicit, content-specific evidence, such as a direct quote, carries significantly more weight than ambiguous, general evidence, like a vague facial expression. Furthermore, the *Primacy of Verbal Content* gives precedence to a subject’s own utterances when assessing complex cognitive-emotional states, especially when visual cues are equivocal. Finally, the *Confidence-Informed Weighting* principle ensures that evidence extracted with high confidence from the perception modules is weighted more heavily during reasoning.

Fine-Grained and Interpretability Analysis. Table 1 presents a comprehensive performance comparison. Three key conclusions can be drawn. First, on FG-Score, which best reflects fine-grained recognition, Affective-CoT (0.5805) not only surpasses all open-source E2E models but also achieves highly competitive performance against powerful closed-source models like GPT-4o and Gemini 2.5 Flash-Lite. This validates the effectiveness of our “expert perception + expert reasoning” architecture for specialized, complex tasks. Second, the most significant divergence lies in the interpretability metrics. Our method achieves an EGS of 0.92 and a CRR of 0.86, creating a substantial gap with all baselines. This

Table 1: Comprehensive performance comparison on fine-grained emotion understanding and interpretability. FG-Score is The official MER-FG metric for descriptive accuracy. An asterisk (*) denotes scores reported in original publications. In "Our Variants," -P and -R specify the Perception and Reasoning models used in our framework, respectively.

Type	Model	Method	FG-Score (↑)	EGS (↑)	CRR (↑)
Closed-Source	GPT-4o[19]	Few-Shot	0.5773	0.83	0.71
	Gemini 2.5 Flash-Lite[40]	Few-Shot	0.5823	0.89	0.82
Open-Source	AffectGPT[27]	Publication score[24]	0.4686*	-	-
	PandaGPT[38]	Few-Shot	0.5592	0.67	0.61
	Chat-UniVi [21]	Publication score[24]	0.3362*	0.57	0.37
	Video-LLaVA [31]	Publication score[24]	0.1979*	0.35	0.36
	MiniGPT4-Video [2]	Few-Shot	0.5119	0.67	0.63
	LLaVA-OneVision [22]	Few-Shot	0.5672	0.71	0.78
	Qwen2-VL [42]	Few-Shot	0.5684	0.78	0.65
	Qwen2.5-VL [3]	Few-Shot	0.5694	0.85	0.77
	InternVL2 [7]	Few-Shot	0.5719	0.82	0.78
	Video-LLaMA 2[9]	Few-Shot	0.5361	0.73	0.63
	LLaVA-NeXT[33]	Few-Shot	0.5702	0.84	0.72
Our Variants	Affective-CoT (InternVL-P[7])	Few-Shot CoT	0.5761	0.90	0.83
	Affective-CoT (SALMONN-P[39])	Few-Shot CoT	0.5717	0.88	0.81
	Affective-CoT (Llama3-R[14])	Few-Shot CoT	0.5786	0.91	0.85
	Affective-CoT (DeepSeek-V3[32])	Few-Shot	0.5793	0.91	0.85
	Affective-CoT (DeepSeek-R1[16])	Few-Shot CoT	0.5805	0.92	0.86

Table 2: Categorical alignment evaluation on MER-SEMI.

Model	Acc & W-F1 Avg.
Emotion-LLaMA[8]	0.7308
AffectGPT[27]	0.7497
Affective-CoT (w/o Workflow)	0.7595
Affective-CoT (Full)	0.7600

Table 3: Ablation study of core components.

Model Type	FG-Score (↑)	EGS (↑)	CRR (↑)
Affective-CoT	0.5805	0.92	0.86
w/o Cognitive Workflow	0.5697	0.61	0.25
w/o Structured Evidence	0.5711	0.75	0.68

decisively demonstrates that while other models generate plausible descriptions, only our framework produces truly trustworthy explanations that are faithful to evidence and can resolve modal conflicts. Third, comparing our framework variants reveals that replacing perception or reasoning modules with alternatives (e.g., InternVL[7], Llama-3[14]) leads to performance drops, validating the efficacy of our current Qwen2.5-VL-Perception + DeepSeek-R1-Reasoning[16] combination. We note that the workflow’s marginal gain on this metric is expected, as this coarse-grained evaluation does not fully capture its primary role in resolving subtle cross-modal conflicts—a contribution decisively validated by the catastrophic drop in interpretability metrics shown in our ablation study.

Ablation Study. To verify the necessity of our core design choices, we conduct an ablation study on two key components of Affective-CoT, with results summarized in Table 3. First, removing the **Cognitive Workflow** ('w/o Workflow') causes a catastrophic

decline in interpretability: EGS and CRR plummet by 33.7% and 70.9%, respectively, despite only a moderate drop in FG-Score. This stark difference empirically proves that our structured workflow, while having a subtle impact on the final descriptive accuracy, is indispensable for generating faithful and conflict-aware explanations. Furthermore, replacing structured JSON evidence with free-form text ('w/o Structured Evidence') leads to a consistent degradation across all metrics. This confirms that providing the reasoning module with structured, symbolic evidence is crucial for it to perform subsequent reasoning with precision and efficiency.

5 Conclusion

In this paper, we introduced Affective-CoT, a novel hierarchical framework that significantly enhances the interpretability and faithfulness of Multimodal Emotion Recognition by explicitly decoupling perception from reasoning. Our extensive experiments, validated by securing first place in the MER-2025 DES human evaluation challenge, demonstrate a critical finding: Affective-CoT achieves state-of-the-art descriptive accuracy on par with leading opaque models, while uniquely providing a transparent, evidence-grounded reasoning process. This work champions a paradigm shift from end-to-end systems towards scrutable, compositional reasoning, representing a vital step for deploying affective computing in high-stakes domains. We acknowledge that the multi-stage nature of our framework introduces computational overhead. Future research will therefore focus on mitigating this limitation by exploring lightweight architectures, such as knowledge distillation for the reasoning module, to enable broader applications without sacrificing its core principles of trustworthiness. Ultimately, Affective-CoT paves the way for building more reliable and auditable AI systems that users can truly understand and trust.

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